Electrical Circuits 1 (DC)

Lecture 2: Voltage and Current

Reference Books

Introductory Circuit Analysis

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OBJECTIVES

- Become aware of the basic *atomic structure of conductors such as copper and aluminum* and understand why they are used so extensively in the field.

- Understand how the terminal *voltage of a battery or any dc supply* is established and *how it creates a flow of charge* in the system.

- Understand *how current is established in a circuit* and how its magnitude is affected by the charge flowing in the system and the time involved.

- Become familiar with the factors that affect the terminal voltage of a battery and how long a battery will remain effective.

- Be able to *apply a voltmeter and ammeter correctly* to measure the voltage and current of a network.

ATOMS AND THEIR STRUCTURE

- **Nucleus**
  - Protons
  - Electrons
  - Neutrons

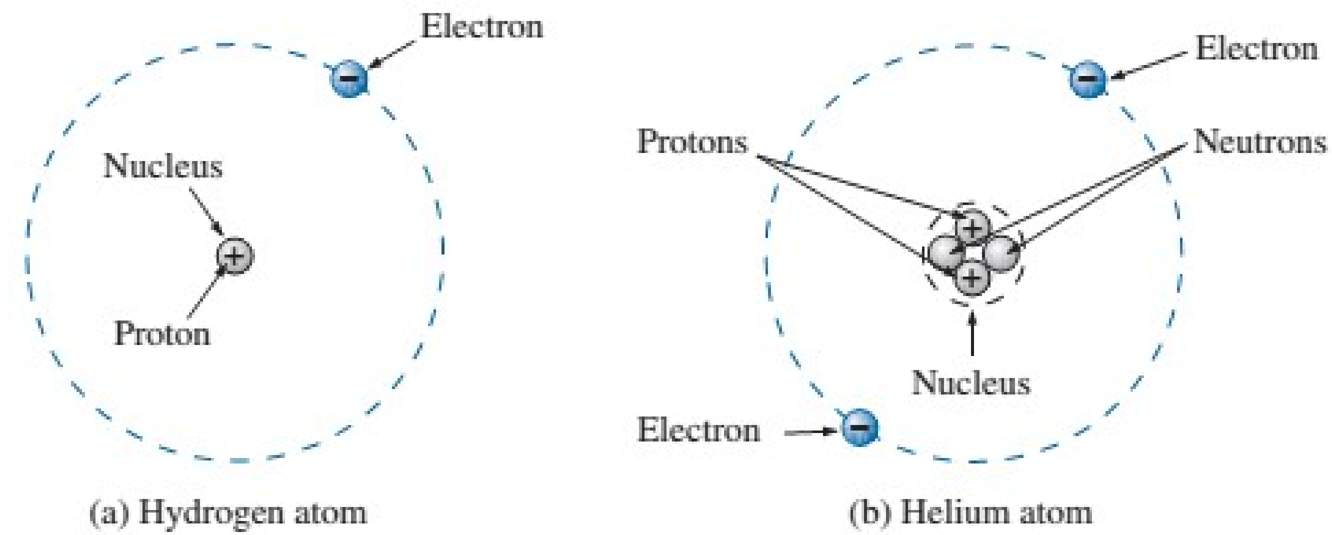

FIG. 2.1
Hydrogen and helium atoms.

a) The orbiting electron carries a negative charge equal in magnitude to the positive charge of the proton.
b) The atomic structure of any stable atom has an equal number of electrons and protons.

ATOMS AND THEIR STRUCTURE

- Copper is the most commonly used metal in the electrical/electronics industry.

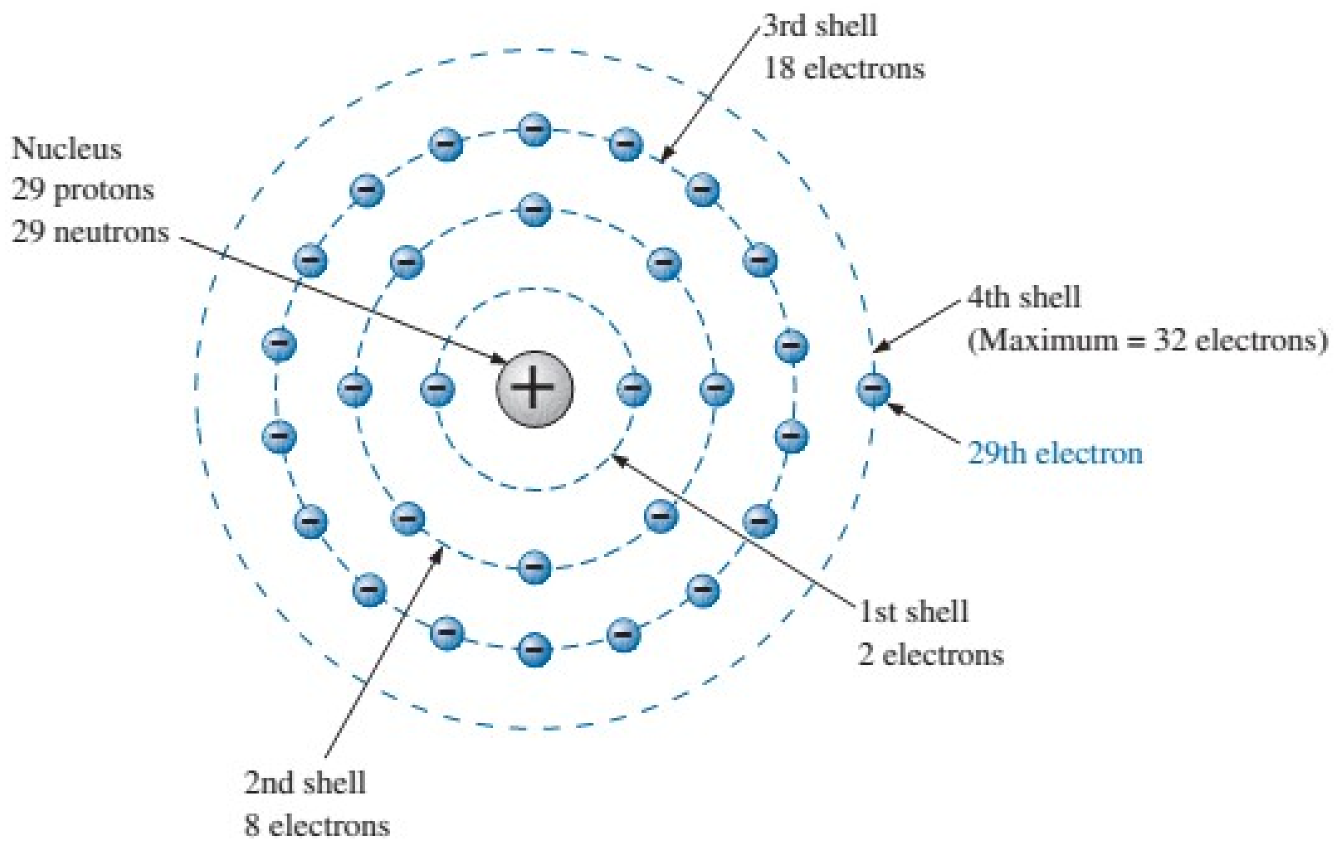

FIG. 2.2
The atomic structure of copper.

VOLTAGE

- The flow of charge is established by an external “pressure” derived from the energy that a mass has by virtue of its position: Potential energy.

- **Energy: the capacity to do work**

- If a mass (m) is raised to some height (h) above a reference plane, it has a measure of potential energy expressed in joules (J) that is determined by

$$\mathbf{W\ (potential\ energy)} = \boldsymbol{mgh}$$

where g is the gravitational acceleration ($9.8\ m/s^2$)

- A potential difference of 1 volt (V) exists between two points if 1 joule (J) of energy is exchanged in moving 1 coulomb (C) of charge between the two points.

- The unit of measurement volt was chosen to honor Alessandro Volta.

VOLTAGE

- A potential difference or voltage is always measured between two points in the system. Changing either point may change the potential difference between the two points under investigation.
- Potential difference between two points is determined by:

$$V = W/Q \quad \text{(volts)}$$

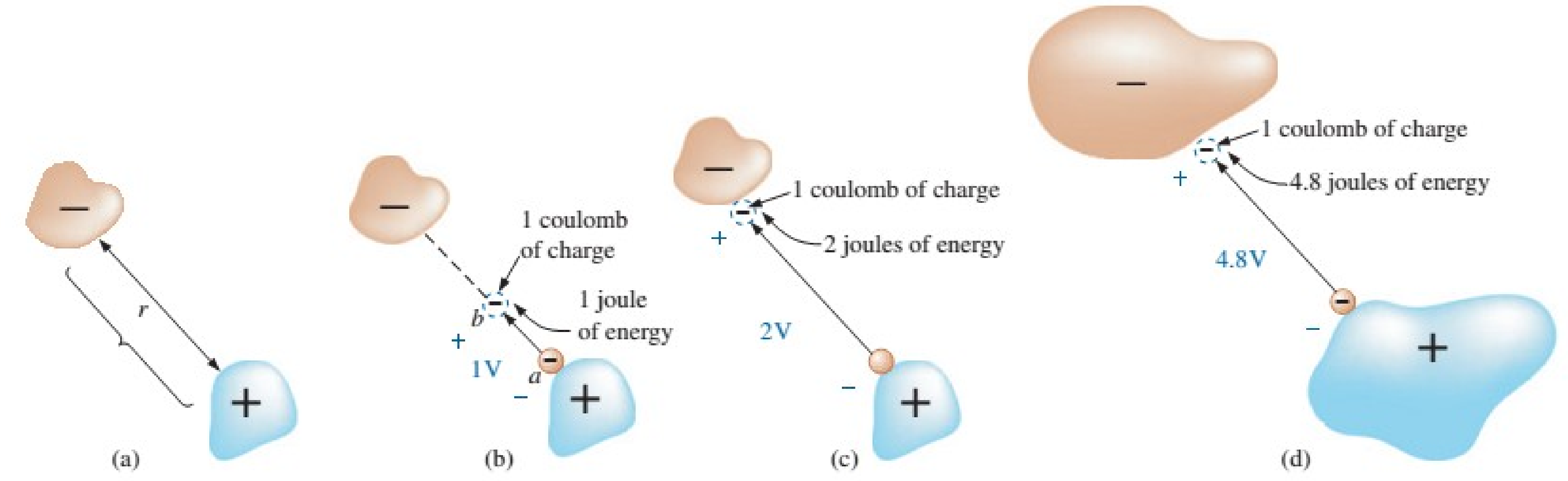

FIG. 2.5
Defining the voltage between two points.

VOLTAGE

- **Potential difference:** The algebraic difference in potential (or voltage) between two points of a network.
- **Voltage:** When isolated, like potential, the voltage at a point with respect to some reference such as ground.
- **Voltage difference:** The algebraic difference in voltage (or potential) between two points of a system. A voltage drop or rise is as the terminology would suggest.
- **Electromotive force (emf):** The force that establishes the flow of charge (or current) in a system due to the application of a difference in potential.

CURRENT

- The free electron is the charge carrier in a copper wire or any other solid conductor of electricity.
- With no external forces applied, the net flow of charge in a conductor in any one direction is zero.
- The applied voltage is the starting mechanism—the current is a reaction to the applied voltage.

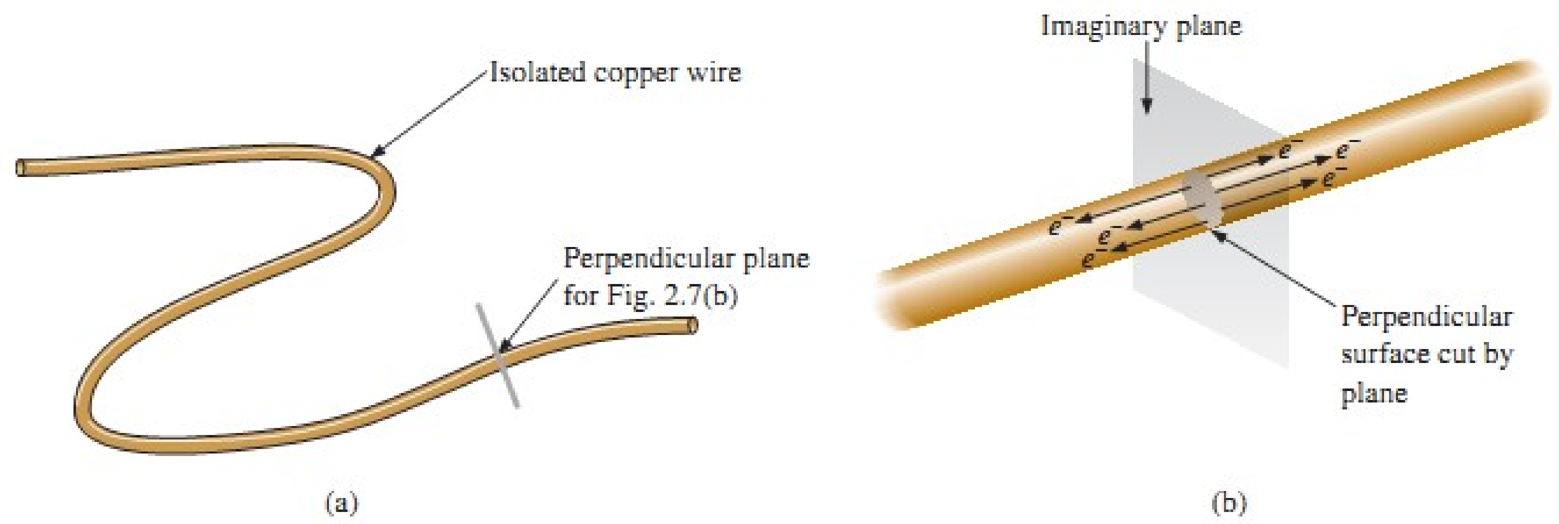

FIG. 2.7
There is motion of free carriers in an isolated piece of copper wire, but the flow of charge fails to have a particular direction.

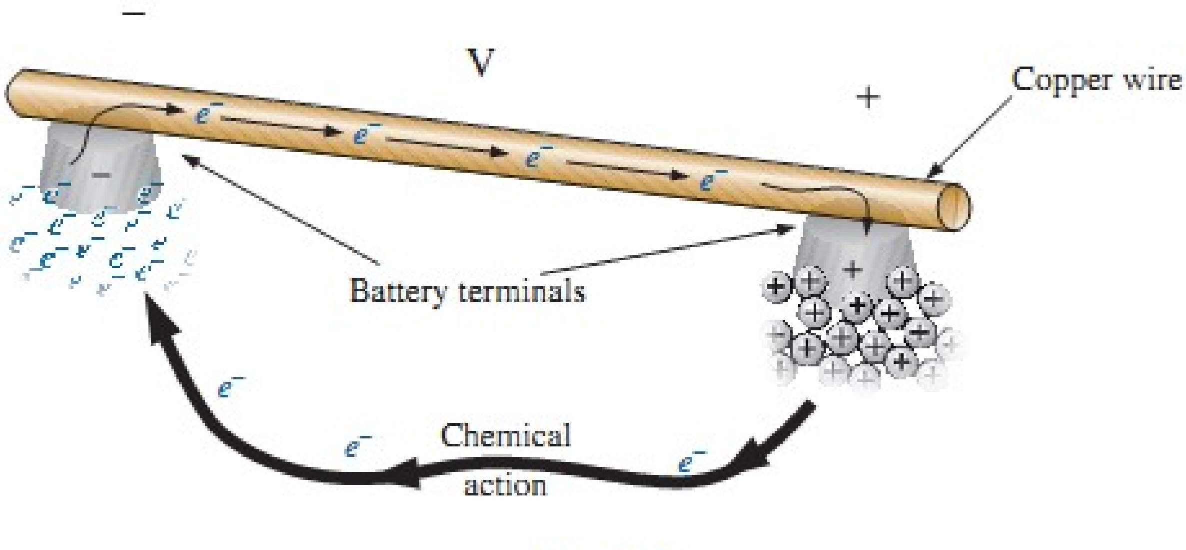

FIG. 2.8
Motion of negatively charged electrons in a copper wire when placed across battery terminals with a difference in potential of volts (V).

CURRENT

The applied voltage (or potential difference) in an electrical/electronics system is the "pressure" to set the system in motion, and the current is the reaction to that pressure.

if 6.242*(10^18) electrons (1 coulomb) pass through the imaginary plane in Fig. 2.9 in 1 second, the flow of charge, or current, is said to be 1 ampere (A).

$$\boxed{I = \frac{Q}{t}}$$

I = amperes (A)
Q = coulombs (C)
t = time (s)

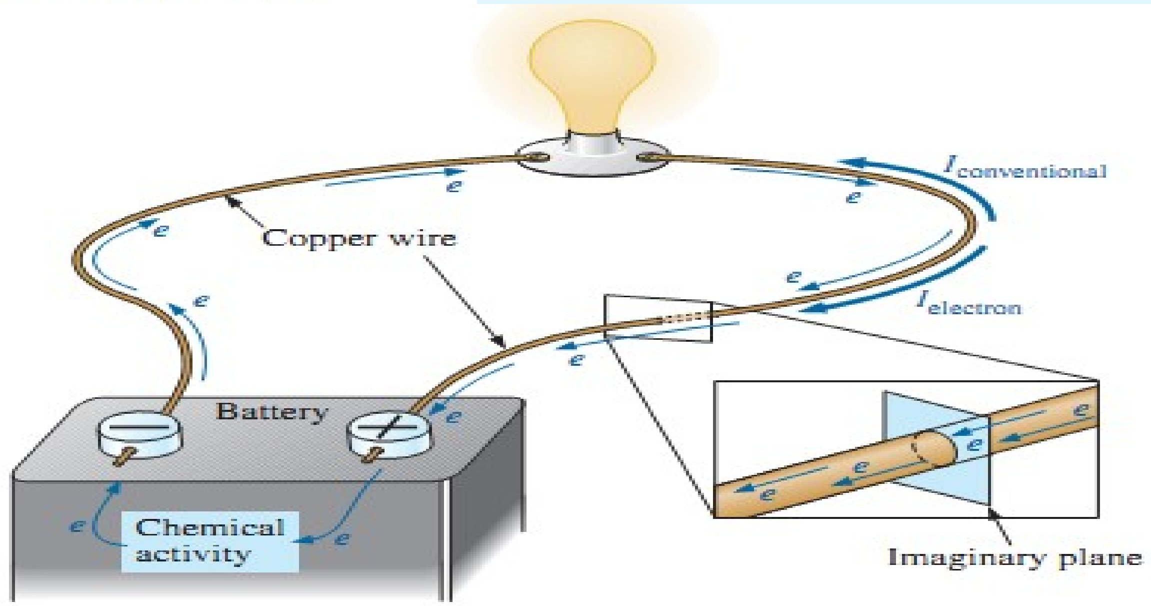

FIG. 2.9
Basic electric circuit.

VOLTAGE SOURCES

- **dc – Direct current:**
  - Unidirectional (“one direction”) flow of charge
  - Supplies that provide a fixed voltage or current

- **dc Voltage sources**
  - Batteries (chemical action)
  - Generators (electromechanical)
  - Power supplies (rectification)

- ***An electromotive force (emf)*** is a force that establishes the flow of charge (or current) in a system due to the application of a difference in potential.

VOLTAGE SOURCES

- **Batteries: combination of two or more similar cells**
  - A cell being a fundamental source of electrical energy developed through the conversion of chemical or solar energy
  - All cells are divided into Primary and Secondary types
    - Primary type is not rechargeable
    - Secondary is rechargeable; the cell can be reversed to restore its capacity
    - Two most common rechargeable batteries are the lead-acid unit (primarily automotive) and the nickel-cadmium (calculators, tools, photoflash units and shavers)

VOLTAGE SOURCES

- **Solar cell:**
  - A fixed illumination of the solar cell will provide a fairly steady dc voltage for driving loads from watches to automobiles
  - Conversion efficiencies are currently between 10% and 14%

- **Ampere-hour rating**
  - Batteries have a capacity rating in ampere-hours
  - A battery with an ampere-hour rating of 100 will theoretically provide a steady current of 1A for 100 h, 2A for 50 h or 10A for 10 h
  - Factors affecting the rating: rate of discharge and temperature
    - The capacity of a dc battery decreases with an increase in the current demand
    - The capacity of a dc battery decreases at relatively (compared to room temperature) low and high temperatures

CONDUCTORS AND INSULATORS

- Conductors are those materials that permit a generous flow of electrons with very little external force (voltage) applied

In addition,

- Good conductors typically have only one electron in the valance (most distant from the nucleus) ring.
- Insulators are those materials that have very few free electrons and require a large applied potential (voltage) to establish a measurable current level
- Insulators are commonly used as covering for current-carrying wire, which, if uninsulated, could cause dangerous side effects
- Rubber gloves and rubber mats are used to help insulated workers when working on power lines
- Even the best insulator will break down if a sufficiently large potential is applied across it

SEMICONDUCTORS

- Semiconductors are a specific group of elements that exhibit characteristics between those of insulators and conductors.
- Semiconductor materials typically have four electrons in the outermost valence ring.
- Semiconductors are further characterized as being photoconductive and having a negative temperature coefficient.
  - Photoconductivity: Photons from incident light can increase the carrier density in the material and thereby the charge flow level
  - Negative temperature coefficient: Resistance will decrease with an increase in temperature (opposite to that of most conductors)

AMMETERS AND VOLTMETERS

- **Ammeter (Milliammeter or Microammeter)**
  - Used to measure current levels
  - Must be placed in the network such that the charge will flow through the meter
- **Voltmeter**
  - Used to measure the potential difference between two points
- **Volt-ohm-milliammeter (VOM) and digital multimeter (DMM):**
  - Both instruments will measure voltage and current and a third quantity, resistance
  - The VOM uses an analog scale, which requires interpreting the position of the pointer on a continuous scale
  - The DMM provides a display of numbers with decimal point accuracy determined by the chosen scale.

APPLICATIONS

- **Flashlight**
  - Simplest of electrical circuits
  - Batteries are connected in series to provide a higher voltage (sum of the battery voltages)
- **12-V Car battery charger**
  - Used to convert 120-V ac outlet power to dc charging power for a 12-V automotive battery, using a transformer to step down the voltage, diodes to rectify the ac (convert it to dc), and in some cases a regulator to provide a dc voltage that varies with level of charge.
- **Answering machines/Phones dc supply**
  - A wide variety of devices receive their dc operating voltage from an ac/dc conversion system
  - The conversion system uses a transformer to step the voltage down to the appropriate level, then diodes “rectify” the ac to dc, and capacitors provide filtering to smooth out the dc.

END